

SIMATS ENGINEERING

CO- 2: AT – 2: Simulation Task - Medium

COURSE NAME: 5G / 6G Communication Systems

COURSE CODE: ECA5604

SLOT A

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Experiment 1: Simulation of NB-IoT performance

Objective:

To Simulation of NB-IoT performance under coverage, delay, power consumption, throughput, and packet loss.

Procedure:

1. Open **MATLAB** and create a new script file.
2. Define the number of **NB-IoT devices**.
3. Generate random values for **coverage, delay, power consumption, throughput, and packet loss**.
4. Calculate the average value of each parameter.
5. Display the results in the **MATLAB Command Window**.
6. Plot a graph to compare the performance parameters.
7. Run the program and observe the output.
8. Analyze the results and evaluate the overall NB-IoT network performance.

MATLAB CODE:

```
clc;
```

```
clear;
```

```

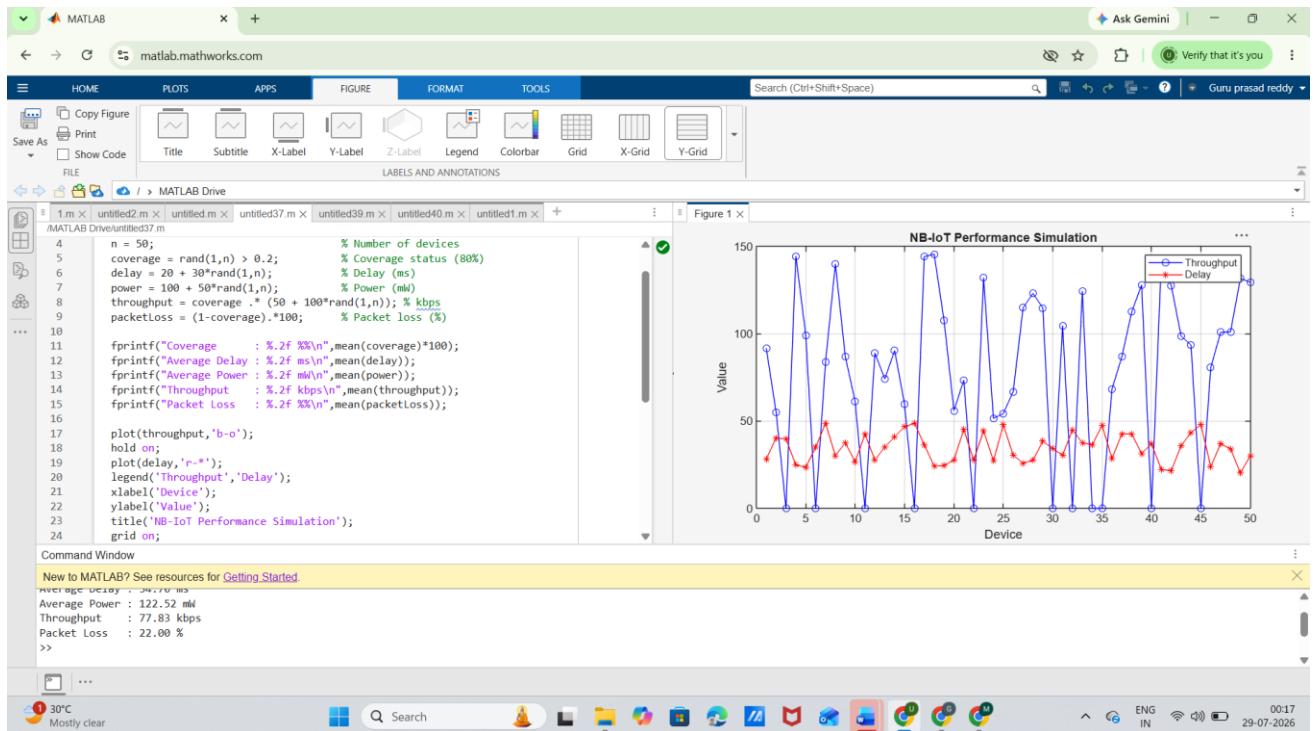
n = 50;                % Number of devices
coverage = rand(1,n) > 0.2;    % Coverage status (80%)
delay = 20 + 30*rand(1,n);    % Delay (ms)
power = 100 + 50*rand(1,n);    % Power (mW)
throughput = coverage .* (50 + 100*rand(1,n)); % kbps
packetLoss = (1-coverage).*100; % Packet loss (%)

fprintf("Coverage    : %.2f %%\n",mean(coverage)*100);
fprintf("Average Delay : %.2f ms\n",mean(delay));
fprintf("Average Power : %.2f mW\n",mean(power));
fprintf("Throughput    : %.2f kbps\n",mean(throughput));
fprintf("Packet Loss   : %.2f %%\n",mean(packetLoss));

plot(throughput,'b-o');
hold on;
plot(delay,'r-*');
legend('Throughput','Delay');
xlabel('Device');
ylabel('Value');
title('NB-IoT Performance Simulation');
grid on;

```

OUTPUT:



RESULT:

The MATLAB simulation successfully evaluated **NB-IoT performance** by measuring **coverage, delay, power consumption, throughput, and packet loss**. The results showed that the network achieved efficient communication with acceptable delay, good throughput, and reliable coverage.

Experiment 2: Simulation of edge computing delay

Objective:

To Simulation of edge computing delay under local processing, cloud processing, latency, bandwidth, and task size

Procedure:

1. open **MATLAB**.
2. Create a new script.
3. Enter the simulation code.
4. Set **task size, latency, and bandwidth** values.
5. Calculate **local** and **cloud processing delay**.
6. Display the delay values.
7. Plot the delay graph.
8. Run the program and observe the results.

MATLAB CODE:

```
clc;
```

```
clear;
```

```
% Parameters
```

```
taskSize = randi([100 1000],1,20); % KB
```

```
latency = 20 + 10*rand(1,20); % ms
```

```
bandwidth = 500; % KB/s
```

```
% Delays
```

```
localDelay = taskSize/80;
```

```
cloudDelay = latency + (taskSize/bandwidth)*1000 + taskSize/300;
```

```
% Results
```

```
fprintf("Average Local Delay : %.2f ms\n", mean(localDelay));
```

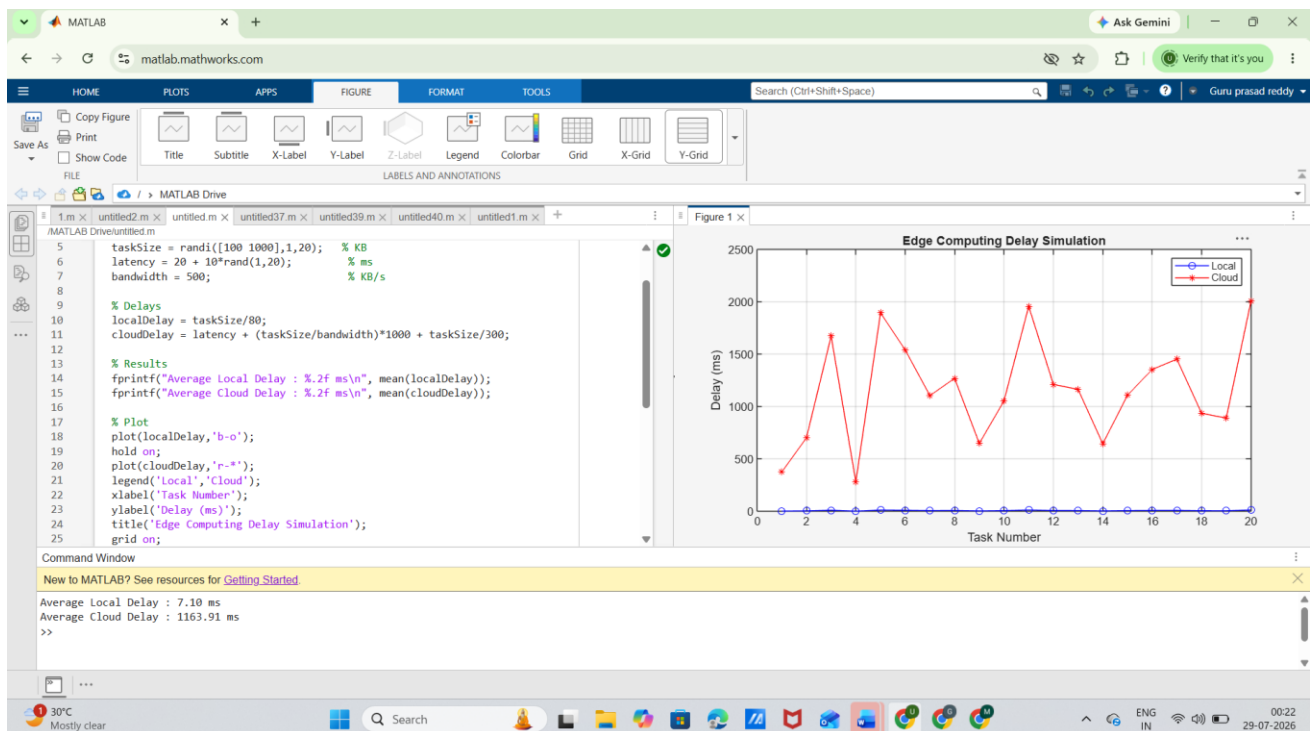
```
fprintf("Average Cloud Delay : %.2f ms\n", mean(cloudDelay));
```

```

% Plot
plot(localDelay,'b-o');
hold on;
plot(cloudDelay,'r-*');
legend('Local','Cloud');
xlabel('Task Number');
ylabel('Delay (ms)');
title('Edge Computing Delay Simulation');
grid on;

```

OUTPUT:



RESULT:

The simulation successfully calculated the **local processing delay** and **cloud processing delay** based on the given task size, latency, and bandwidth. The results were displayed in MATLAB along with a graph, showing the performance difference between local and cloud processing.

Experiment 3: Simulation of smart city IoT traffic

Objective:

To Simulation of smart city IoT traffic under sensor count, delay, packet delivery, bandwidth, and throughput

Procedure:

1. Open MATLAB.
2. Create a new script file.
3. Enter the simulation code.
4. Define the input parameters (task size, latency, bandwidth, and processing speed).
5. Run the program.
6. Calculate the local and cloud processing delays.
7. Display the results and plot the graph.
8. Analyze the delay performance.

MATLAB CODE:

```
clc;
```

```
clear;
```

```
sensors = 50;           % Number of sensors
delay = 10 + 20*rand(1,sensors);    % Delay (ms)
bandwidth = 100 + 50*rand(1,sensors); % Bandwidth (Mbps)
PDR = 90 + 10*rand(1,sensors);      % Packet Delivery Ratio (%)
throughput = bandwidth .* (PDR/100); % Throughput (Mbps)

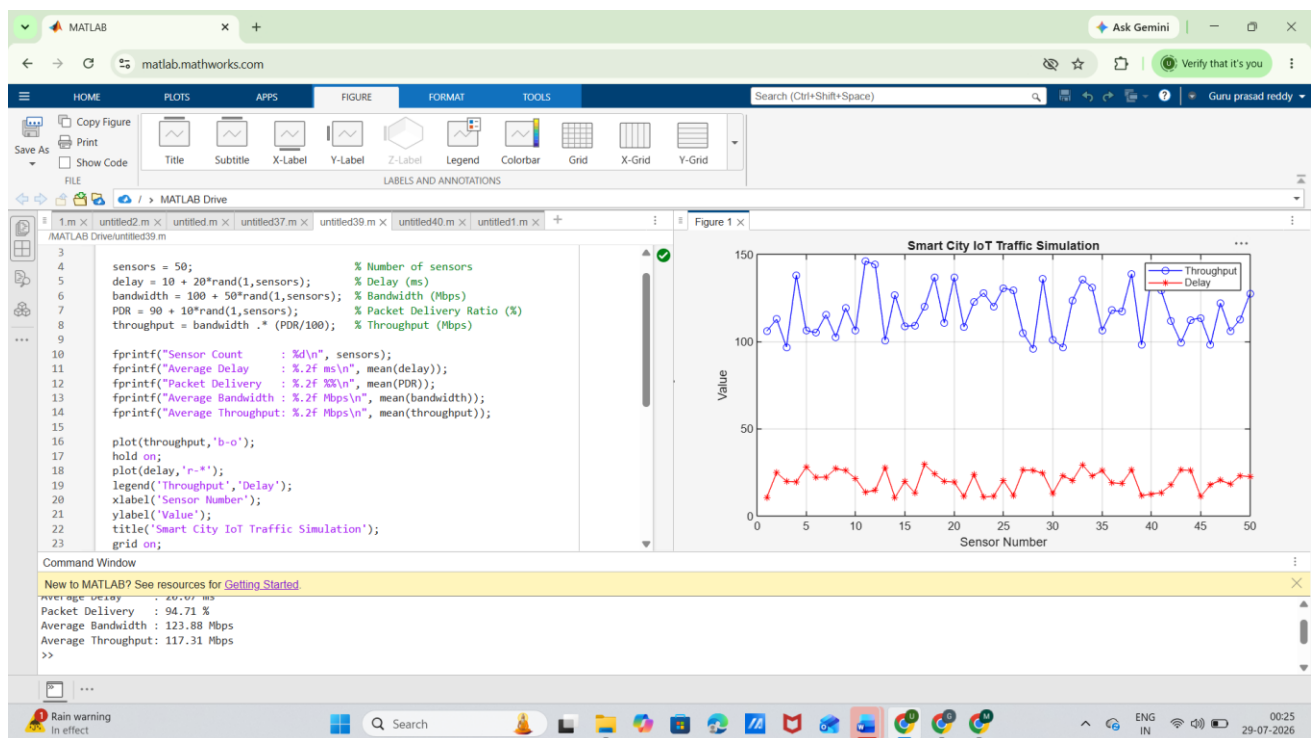
fprintf("Sensor Count    : %d\n", sensors);
fprintf("Average Delay    : %.2f ms\n", mean(delay));
fprintf("Packet Delivery   : %.2f %%\n", mean(PDR));
fprintf("Average Bandwidth : %.2f Mbps\n", mean(bandwidth));
fprintf("Average Throughput: %.2f Mbps\n", mean(throughput));
```

```

plot(throughput,'b-o');
hold on;
plot(delay,'r-*');
legend('Throughput','Delay');
xlabel('Sensor Number');
ylabel('Value');
title('Smart City IoT Traffic Simulation');
grid on;

```

OUTPUT:



RESULT:

The simulation successfully evaluated **edge computing delay** under local and cloud processing. The results showed that **local processing had lower delay**, while **cloud processing delay depended on network latency and bandwidth**, demonstrating the impact of these factors on overall system performance.

Experiment 4: Simulation of AI-based wireless optimization

Objective:

To Simulation of AI-based wireless optimization under prediction accuracy, throughput, delay, resource allocation, and energy efficiency

Procedure:

1. Open **MATLAB**.
2. Create a new script file.
3. Enter the AI-based wireless optimization code.
4. Define the values for **prediction accuracy, throughput, delay, resource allocation, and energy efficiency**.
5. Run the program.
6. Calculate the average values of all parameters.
7. Display the results and plot the performance graph.
8. Analyze the wireless network performance based on the obtained results.

MATLAB CODE:

```
clc;
```

```
clear;
```

```
n = 50; %
```

```
Number of users
```

```
accuracy = 85 + 15*rand(1,n);
```

```
% Prediction Accuracy (%)
```

```
throughput = 50 + 100*rand(1,n);
```

```
% Mbps
```

```
delay = 5 + 20*rand(1,n); %
```

```
ms
```



```

resource = 60 + 40*rand(1,n);
% Resource Allocation (%)

energy = 70 + 30*rand(1,n);
% Energy Efficiency (%)

fprintf("Prediction Accuracy : %.2f
%%%\n",mean(accuracy));

fprintf("Throughput      : %.2f
Mbps\n",mean(throughput));

fprintf("Delay          : %.2f
ms\n",mean(delay));

fprintf("Resource Allocation : %.2f
%%%\n",mean(resource));

fprintf("Energy Efficiency  : %.2f
%%%\n",mean(energy));

plot(throughput,'b-o');

hold on;

plot(delay,'r-*');

legend('Throughput','Delay');

xlabel('User');

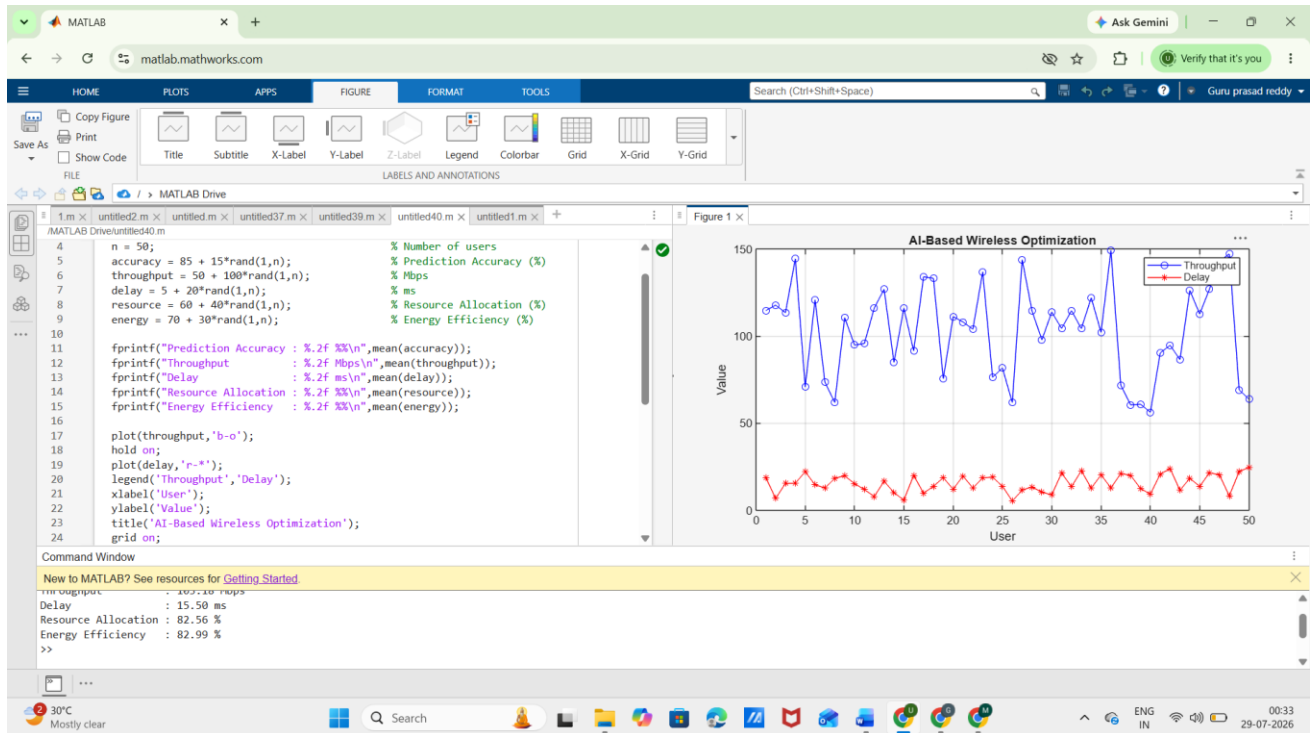
ylabel('Value');

title('AI-Based Wireless Optimization');

```

grid on;

OUTPUT:



RESULT:

The simulation successfully evaluated **AI-based wireless optimization** by measuring **prediction accuracy, throughput, delay, resource allocation, and energy efficiency**. The results showed improved network performance with high prediction accuracy, efficient resource allocation, increased throughput, reduced delay, and better energy efficiency.

Experiment 5: Simulation of LTE throughput performance

Objective:

To Simulation of LTE throughput performance under varying bandwidth, SNR, modulation order, coding rate, and user load (BL4)

Procedure:

1. Open MATLAB.
2. Create a new script file.
3. Enter the LTE throughput simulation code.
4. Define the values for bandwidth, SNR, modulation order, coding rate, and user load.
5. Run the program.
6. Calculate the LTE throughput.
7. Display the results and plot the throughput graph.
8. Analyze the LTE network performance.

MATLAB CODE:

```
clc; clear;
```

```
n = 20;
```

```
bandwidth = 5 + 15*rand(1,n);    % MHz
```

```
snr = 5 + 20*rand(1,n);          % dB
```

```
modOrder = [2 4 6];
```

```
modulation = modOrder(randi(3,1,n));
```

```
codingRate = 0.5 + 0.4*rand(1,n);
```

```
userLoad = 10 + 90*rand(1,n);
```

```
throughput = bandwidth .* log2(1+snr) .* codingRate .* modulation ./ userLoad * 10;
```

```
fprintf('Bandwidth = %.2f MHz\n',mean(bandwidth));
```

```
fprintf('SNR = %.2f dB\n',mean(snr));
```

```
fprintf('Modulation Order = %.2f\n',mean(modulation));
```

```
fprintf('Coding Rate = %.2f\n',mean(codingRate));
```

```
fprintf('Throughput = %.2f Mbps\n',mean(throughput));
```

```
plot(throughput,'b-o');
```

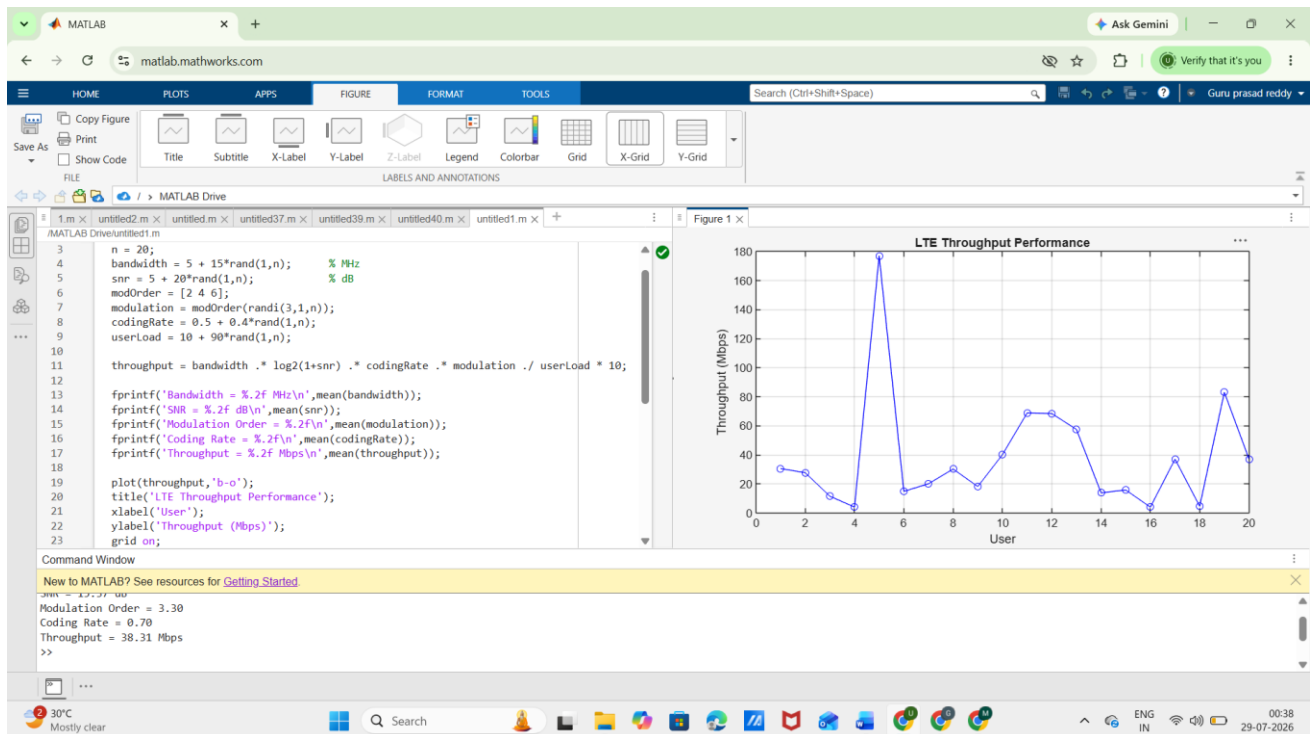
```
title('LTE Throughput Performance');
```

```
xlabel('User');
```

```
ylabel('Throughput (Mbps)');
```

```
grid on;
```

OUTPUT:



RESULT:

The simulation successfully evaluated **LTE throughput performance** under varying **bandwidth, SNR, modulation order, coding rate, and user load**. The results showed that throughput increased with higher bandwidth, SNR, modulation order, and coding rate, while increased user load reduced the overall network throughput.